

DPP NO.-2

- Q.1 How many of the 900 three digit numbers have at least one even digit?
(A) 775 (B) 875 (C) 450 (D) 750
- Q.2 The number of natural numbers from 1000 to 9999 (both inclusive) that do not have all 4 different digits is
(A) 4048 (B) 4464 (C) 4518 (D) 4536
- OR**
- What can you say about the number of even numbers under the same constraints?
- Q.3 The number of different seven digit numbers that can be written using only three digits 1, 2 & 3 under the condition that the digit 2 occurs exactly twice in each number is :
(A) 672 (B) 640 (C) 512 (D) none
- Q.4 Out of seven consonants and four vowels, the number of words of six letters, formed by taking four consonants and two vowels is (Assume that each ordered group of letter is a word):
(A) 210 (B) 462 (C) 151200 (D) 332640
- Q.5 All possible three digits even numbers which can be formed with the condition that if 5 is one of the digit, then 7 is the next digit is :
(A) 5 (B) 325 (C) 345 (D) 365
- Q.6 For some natural N , the number of positive integral ' x ' satisfying the equation ,
 $1! + 2! + 3! + \dots + (x!) = (N)^2$ is :
(A) none (B) one (C) two (D) infinite
- Q.7 The number of six digit numbers that can be formed from the digits 1, 2, 3, 4, 5, 6 & 7 so that digits do not repeat and the terminal digits are even is :
(A) 144 (B) 72 (C) 288 (D) 720
- Q.8 A new flag is to be designed with six vertical strips using some or all of the colours yellow, green, blue and red. Then, the number of ways this can be done such that no two adjacent strips have the same colour is
(A) 12×81 (B) 16×192 (C) 20×125 (D) 24×216
- Q.9 In how many ways can 5 colours be selected out of 8 different colours including red, blue, and green
(a) if blue and green are always to be included,
(b) if red is always excluded,
(c) if red and blue are always included but green excluded?
- Q.10 A 5 digit number divisible by 3 is to be formed using the numerals 0, 1, 2, 3, 4 & 5 without repetition. The total number of ways this can be done is :
(A) 3125 (B) 600 (C) 240 (D) 216
- Q.11 Number of 9 digits numbers divisible by nine using the digits from 0 to 9 if each digit is used atmost once is $K \cdot 8!$, then K has the value equal to _____.
- Q.12 Number of natural numbers less than 1000 and divisible by 5 can be formed with the ten digits, each digit not occuring more than once in each number is _____.